

ADVANCES IN HEART FAILURE, MECHANICAL CIRCULATORY SUPPORT AND TRANSPLANT

A Recovery-Focused Framework for the Management of Cardiogenic Shock

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ABSTRACT: Cardiogenic shock describes a syndrome of systemic hypoperfusion due to insufficient cardiac output. Therapeutic interventions aim to restore systemic perfusion and oxygen delivery, relieve congestion, and reverse the primary culprit pathologies that precipitate shock. Cardiogenic shock care often emphasizes acute rescue and survival; however, survival does not always equal meaningful recovery. In this review, we propose a recovery-focused framework in which myocardial and end-organ recovery are treated as explicit, measurable targets. Building upon prior frameworks proposed for the management of shock, we organize cardiogenic shock management into four overlapping phases: salvage, optimization, stabilization, and de-escalation. Each phase addresses a distinct clinical question: can the patient be rescued, is perfusion and unloading sufficient to prevent further injury, is meaningful myocardial and end-organ recovery occurring, and is recovery durable enough to withdraw support? We synthesize current strategies and highlight emerging targets and research priorities spanning early identification and triage, integrated macro- and microcirculatory monitoring, phenotype-guided management, targeted metabolic and immunologic therapies, use of mechanical circulatory support as a recovery platform, and structured de-escalation of care.

Key Words: heart failure ■ heart-assist devices ■ hemodynamic monitoring ■ microcirculation ■ myocardial infarction ■ shock, cardiogenic ■ ventricular remodeling

Cardiogenic shock (CS) defines a heterogeneous, multifactorial syndrome characterized by systemic hypoperfusion due to inadequate cardiac output.^{1,2} It can be the result of various pathologies affecting the cardiac muscle, valves, pericardium, conduction system, and systemic and pulmonary vasculature, with downstream effects on peripheral organs.^{2–4} Immediate priorities in contemporary CS care are to restore oxygenation and perfusion, relieve congestion, and reverse the underlying culprit pathologies when possible, in an attempt to stabilize the patient. Stabilization and rescue in the short term, however, do not always lead to long-term myocardial and end-organ recovery, with implications for survival and quality of life.^{2–4} Myocardial recovery, defined as sufficient cardiac functional and structural improvement to permit native heart survival (NHS; ie, survival without the need for advanced heart failure [HF] therapies),^{5–8} accompanied by end-organ recovery and physical and cognitive recuperation, should be an explicit, measurable target of

CS care. While epidemiological data on end-organ recovery and functional status after CS are sparse, single- and multicenter registries suggest that myocardial recovery rates (defined as NHS) vary substantially depending on CS cause. In acute myocardial infarction (AMI)-related CS (AMI-CS), in-hospital mortality has been reported at ≈30% to 40%, but the majority of survivors achieve NHS, as advanced HF therapies are infrequently pursued (≈2.5% in single-center studies).^{7,9–11} In contrast, HF-related CS (HF-CS) is associated with lower in-hospital mortality rates (≈20%–30%), yet a lower proportion of survivors achieves NHS, as ≈10% to 40% of patients undergo advanced HF therapies.^{8–10,12} In the RESTAGE-HF trial (Remission From Stage D Heart Failure), 40% of selected patients supported with a durable left ventricular assist device (LVAD) achieved myocardial recovery sufficient to allow for LVAD weaning.¹³ Multicenter registry data suggest that ≈10% to 15% of consecutive patients with LVAD achieve myocardial recovery to

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Nonstandard Abbreviations and Acronyms

AMI	acute myocardial infarction
AMI-CS	acute myocardial infarction-related cardiogenic shock
CS	cardiogenic shock
FAO	fatty acid oxidation
HF	heart failure
HF-CS	heart failure-related cardiogenic shock
IABP	intraaortic balloon pump
LV	left ventricular
LVAD	left ventricular assist device
MCS	mechanical circulatory support
MCT	monocarboxylic anion transporter
NHS	native heart survival
PVAD	percutaneous ventricular assist device
RV	right ventricular
SCAI	Society for Cardiovascular Angiography and Interventions
VA-ECMO	veno-arterial extracorporeal membrane oxygenation

a degree that LVAD weaning might be considered.^{14–16} In this review, we synthesize contemporary evidence, highlight knowledge gaps, and propose research priorities for the management of CS, where myocardial and end-organ recovery is the central therapeutic objective. We build upon frameworks proposed for the management of circulatory shock and CS^{17,18} by positioning myocardial and end-organ recovery as the central organizing end point, and propose a recovery-focused framework covering early detection and triage, macro- and micro-circulatory monitoring, phenotype-guided management, targeted metabolic and immunologic therapies, use of temporary mechanical circulatory support (MCS), and coordinated de-escalation of care.

RECOVERY-FOCUSED FRAMEWORK FOR CS MANAGEMENT

Management of CS can be conceptualized as four overlapping and dynamic phases (salvage, optimization, stabilization, de-escalation), similar to prior frameworks proposed for the management of circulatory shock and, more recently CS (Figure 1).^{17,18}

1. Salvage phase: Can the patient be rescued? This phase prioritizes immediate life-saving measures. The goal is to achieve a minimum blood pressure and cardiac output that are adequate to sustain survival and end-organ function, with interventions focusing on airway and oxygenation, circulatory support, and correction of potentially reversible causes of the CS (ie, emergent revascularization,

pericardiocentesis, valvular interventions, and other measures). The timeliness of these interventions often determines the extent of downstream organ injury and the likelihood of subsequent recovery.

2. Optimization phase: Is perfusion and unloading sufficient to prevent further injury? This phase focuses on titrating therapies to optimize end-organ perfusion and cardiac unloading, guided by macro- and microcirculatory indices (invasive hemodynamics, end-organ perfusion markers, and other measures). The goal is not only to normalize blood pressure but also to ameliorate the hemometabolic derangements that sustain shock while limiting ongoing cardiac stress.
3. Stabilization phase: Is meaningful myocardial and end-organ recovery occurring? After the initial hemodynamic crisis has been controlled, attention shifts toward mitigating or reversing organ dysfunction. Ventilatory support, renal replacement therapy, correction of coagulopathy, infection control, and prevention and management of complications (vascular, hemorrhagic) are commonly used. Importantly, this phase provides the opportunity to introduce disease-modifying therapies.
4. De-escalation phase: Is recovery durable enough to withdraw support? This final phase entails structured weaning from ventilatory, hemodynamic, and end-organ support. This stage demands vigilant physiological monitoring to assess the degree and sustainability of myocardial recovery, uptitrate disease-modifying therapies, determine if additional time on support may allow further improvement, and determine whether advanced HF therapies or palliation should be considered.

INITIAL EVALUATION AND TRIAGE: THE RECOVERY CLOCK STARTS

Early identification of CS is critical, as it allows for appropriate and timely interventions that influence not only the likelihood of survival but also the potential for myocardial and end-organ recovery.³ The goal is to preserve the reversibility of cardiac and peripheral organ injury by minimizing the time in a low-flow, inflammatory, metabolically stressed state.^{3,19}

Traditional CS definitions emphasize resistant hypotension with evidence of end-organ hypoperfusion, as described in the SHOCK trial (Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock).²⁰ These definitions, however, try to dichotomize the dynamic clinical entity of CS and fail to acknowledge subtypes like normotensive CS, which is increasingly recognized and carries a similarly poor prognosis.²¹ In clinical practice, the diagnosis of CS begins with a suspicion of decreased cardiac output in patients at risk, with typical symptoms and signs including altered mental status,

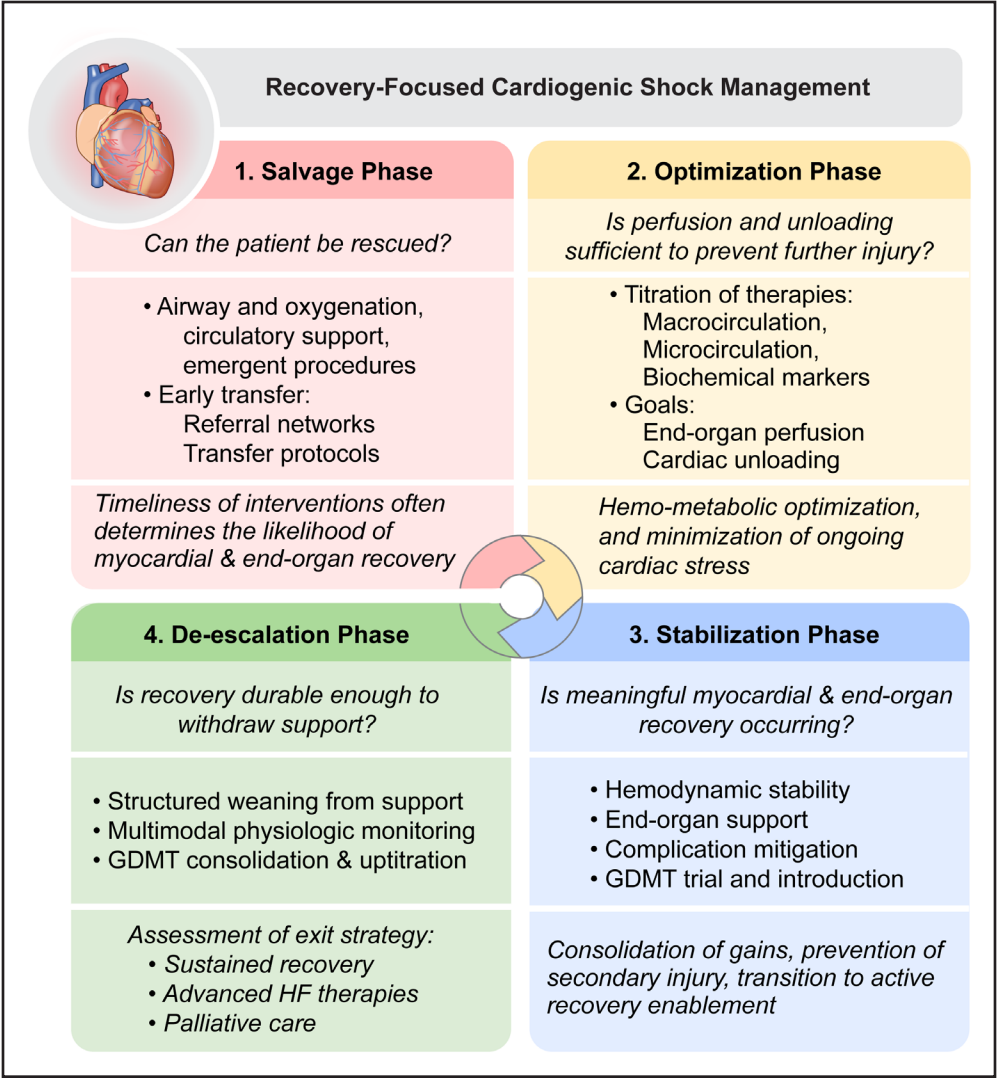


Figure 1. Recovery-focused framework for the management of cardiogenic shock. Four overlapping and dynamic phases include Salvage, Optimization, Stabilization, and De-escalation. GDMT indicates guideline-directed medical therapy; and HF, heart failure.

cold/clammy extremities, tachycardia, narrow pulse pressure, tachypnea and orthopnea, elevated jugular venous pressure, pulmonary congestion, and oliguria or anuria.^{1,4} Laboratory evaluation frequently reveals metabolic acidosis, renal and hepatic dysfunction, elevated serum lactate, and cardiac biomarkers such as troponin and brain-type natriuretic peptide.¹ Central venous oxygen saturation can be helpful.³ Rapid acquisition of a 12-lead ECG and point-of-care echocardiography is essential to identify reversible causes (AMI, cardiac tamponade, severe valvular dysfunction).^{3,4} Invasive hemodynamics are not required for initial diagnosis, although they help elucidate ventricular involvement and guide therapy.¹ Explicit time targets can expedite escalation of care, with the Society for Cardiovascular Angiography and Interventions (SCAI) recently proposing the door-to-lactate clearance (serum lactate <2 mmol/L within 24 hours of diagnosis) as a marker to assess the effectiveness of early management

strategies, and inform the decision to modify or escalate therapies, including transfer to a higher level-of-care institution.²² Multidisciplinary shock teams and referral CS networks have been shown to improve outcomes in single, and multicenter observational studies and could facilitate recovery by allowing earlier access to advanced monitoring and phenotype-specific treatment.^{23–27} Establishing regional referral networks and transfer protocols with clear acceptance criteria could reduce delays and facilitate recovery and NHS.³

INTEGRATED HEMODYNAMIC MONITORING: ALIGNING MACRO- AND MICROCIRCULATORY TARGETS

Treatment strategies in CS are typically titrated toward the more readily apparent and quantifiable macrovascular

hemodynamics, assuming that microvascular hemodynamics will respond in tandem.²⁸ While macrohemodynamic derangements may precipitate the clinical manifestations of CS, the microvascular response and biochemical milieu driven by tissue hypoperfusion, may drive and sustain decompensation.²⁸ As such, effective management requires integrating macrohemodynamics with markers of tissue-level oxygenation, and continuously titrating therapies to facilitate recovery (Figure 2).^{28,29}

Invasive hemodynamic monitoring provides a detailed characterization of cardiac output, left and right-sided filling pressures, and systemic and pulmonary vascular resistance.³ Observational studies suggest that early and complete hemodynamic profiling is associated with improved in-hospital outcomes.^{3,30} Besides the absence of randomized data, contemporary CS populations substantially differ from patients with advanced HF studied in the ESCAPE trial (Evaluation Study of Congestive Heart Failure and Pulmonary Artery Catheterization Effectiveness).^{3,31} Ongoing trials such as the Pulmonary Artery Catheter in CS aim to clarify the role of early invasive hemodynamic assessment (REGISTRATION: URL: <https://clinicaltrials.gov/study/NCT00077948>; Unique identifier: NCT05485376). Standard hemodynamic measurements that have been shown to predict outcomes in CS in observational studies include mean arterial pressure, systemic vascular resistance, pulmonary artery pressures, cardiac index, and disproportionately elevated right ventricular (RV) compared with left ventricular (LV) filling pressures.^{30,32–34} Advanced hemodynamic markers have been introduced to enhance prognostic and therapeutic precision, and their association with outcomes has been validated in observational studies and post hoc analyses of prospective studies.^{33,35} Cardiac power output integrates pressure and flow measurements and is a strong prognostic marker and indicator for escalation of hemodynamic support.^{32,33,36} Aortic pulsatility index quantifies the efficiency of LV ejection and may predict successful weaning from temporary MCS.³⁷ The RV equivalent, pulmonary artery pulsatility index, has been identified as an independent predictor of 30-day mortality.²⁶ RV stroke work index has been shown to be prognostic in HF-CS.³⁸ The myocardial performance score, which combines the aortic pulsatility index and cardiac power output, has been introduced as a metric of ventricular coupling and mechanical LV efficiency.³⁹ Prior investigations have attempted to identify dichotomous cutoffs to define risk categories; however, these metrics exist along a continuum and are best interpreted serially within the dynamic context of CS.²

Microcirculatory indices have been assessed in observational studies and have been shown to prognosticate patients with CS. Capillary refill time is a rapid bedside indicator of peripheral perfusion and was shown to predict adverse outcomes in patients with CS.⁴⁰ The peripheral perfusion index is defined as the ratio of the pulsatile

and nonpulsatile components of the pulse oximetry waveform and provides a noninvasive estimate of microvascular flow.⁴¹ It predicts outcomes in critically ill patients when measured after early resuscitation; however, its accuracy may decrease in low-pulsatility states, which is relevant for patients on MCS.⁴² The direct visualization of microcirculation could serve as a powerful, dynamic tool to assess tissue perfusion, and identify discrepancies between macro- and microhemodynamics.¹⁷ Sublingual video microscopy using sidestream dark-field or incident dark-field imaging represents an established method for direct microcirculatory assessment.^{28,43,44} Handheld devices visualize capillaries in the sublingual area, which serves as an accessible surrogate for splanchnic microvasculature.⁴⁴ Key metrics include the perfused capillary density, proportion of perfused capillaries, and proportion of perfused small vessels, which have been shown to predict outcomes in CS.²⁸ Near-infrared spectroscopy determines tissue oxygen saturation from oxyhemoglobin and deoxyhemoglobin fractions and has been shown to identify patients in more advanced shock.⁴⁵

Serum lactate increases under hypoxic conditions through accelerated glycolysis and impaired mitochondrial oxidation, and serial assessment is a cornerstone in monitoring critically ill patients.^{46,47} Early lactate clearance was a powerful prognostic indicator in secondary analyses of the IABP-SHOCK II (Intraaortic Balloon Pump in Cardiogenic Shock II) and DOREMI (Milrinone as Compared with Dobutamine in the Treatment of Cardiogenic Shock) trials,^{48,49} and the SCAI door-to-lactate clearance initiative emphasizes early normalization as a therapeutic goal.²² Other markers used in the monitoring of patients with CS include mixed venous oxygen saturation, central venous oxygen saturation, and venous-arterial carbon dioxide difference.^{28,29} Mixed venous oxygen saturation and central venous oxygen saturation reflect systemic tissue oxygenation and oxygen extraction and serve as a surrogate for cardiac output.²⁹ The American College of Cardiology recommends serial measurements of mixed venous oxygen saturation or central venous oxygen saturation every 2 to 8 hours depending on the phase of care.³ Venous carbon dioxide is determined by aerobic production of carbon dioxide and, in the presence of normal blood washout, a minimal venous-arterial carbon dioxide difference exists (<2–6 mm Hg).^{29,50} An elevated venous-arterial carbon dioxide difference has been shown to predict adverse outcomes in patients with circulatory failure and CS.^{29,50} Novel biomarkers (dipeptidyl peptidase-3, adrenomedullin, intestinal fatty acid-binding protein, neutrophil gelatinase-associated lipocalin, kidney injury molecule-1, CS 4-protein score, CLIP score, and others) reflect myocardial injury, end-organ dysfunction, and systemic inflammation, and have demonstrated prognostic value in patients with CS in observational studies and post hoc analyses of randomized controlled trials.^{51–55}

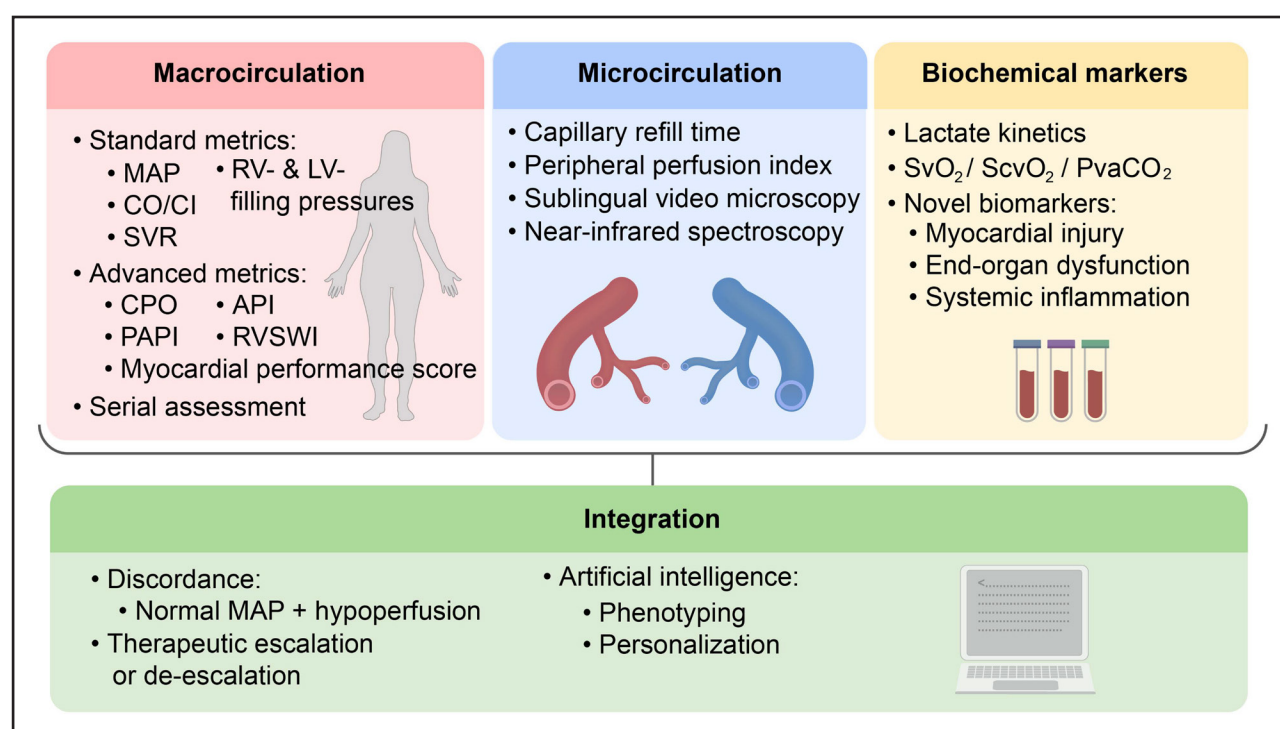


Figure 2. Integrated hemodynamic monitoring: aligning macro- and microcirculatory targets.

API indicates aortic pulsatility index; CI, cardiac index; CO, cardiac output; CPO, cardiac power output; LV, left ventricular; MAP, mean arterial pressure; PAPI, pulmonary artery pulsatility index; PvaCO₂, venous-arterial carbon dioxide difference; RV, right ventricular; RVSWI, right ventricular stroke work index; ScvO₂, central venous oxygen saturation; SvO₂, mixed venous oxygen saturation; and SVR, systemic vascular resistance.

An integrated monitoring strategy combining invasive hemodynamics, tissue perfusion metrics, and biochemical markers, allows clinicians to detect discordance and guide treatment decisions. Given the complexity of multidimensional variables and the heterogeneity of CS, artificial intelligence tools may help clinicians refine phenotyping and inform more personalized, recovery-oriented management.^{56–61}

CLINICAL AND HEMODYNAMIC PHENOTYPING: ONE-SIZE MIGHT NOT FIT ALL

CS defines a dynamic syndrome encompassing diverse etiologies, hemodynamic profiles, and pathophysiologic mechanisms (Figure 3).^{3,4} The phenotype of CS influences the likelihood and time course of myocardial and end-organ recovery and allows clinicians to guide the intensity of support, anticipation of complications, and expectations for NHS.

Cause and Chronicity

CS is broadly categorized into AMI-CS or HF-CS.^{2–4} AMI-CS typically reflects an abrupt ischemic insult due to impaired coronary blood flow, resulting in depressed

cardiac function.^{2–4} It is potentially reversible with rapid revascularization, barring mechanical complications such as papillary muscle rupture leading to mitral regurgitation, ventricular septal defect, or free wall rupture that may require urgent intervention.^{2–4} HF-CS encompasses a heterogeneous spectrum of disease processes and either represents an acute non-AMI insult (de novo HF-CS) or the culmination of progressive myocardial dysfunction (acute-on-chronic HF-CS).^{2,4,9} Underlying etiologies include acute myocarditis (eg, infectious, inflammatory, toxin, or drug-mediated) or various cardiomyopathies (ischemic, arrhythmia-mediated, stress-induced, peripartum, hypertrophic, restrictive, infiltrative, valvular, genetic, idiopathic).^{2–4} Although the above categorization provides a useful clinical framework, a state of pure CS rarely exists. Decreased cardiac output and tissue hypoperfusion frequently trigger systemic inflammatory activation, vasodilation, endothelial dysfunction, and metabolic derangements that further perpetuate shock physiology.^{4,62} Conversely, systemic illnesses can precipitate myocardial dysfunction that evolves into CS. Conditions such as sepsis may initially manifest as distributive shock but can subsequently induce myocardial depression through inflammatory cytokine signaling, mitochondrial dysfunction, and impaired calcium handling, and evolve into a cardiac-dominant phenotype.^{62,63} These bidirectional interactions underscore the complexity of

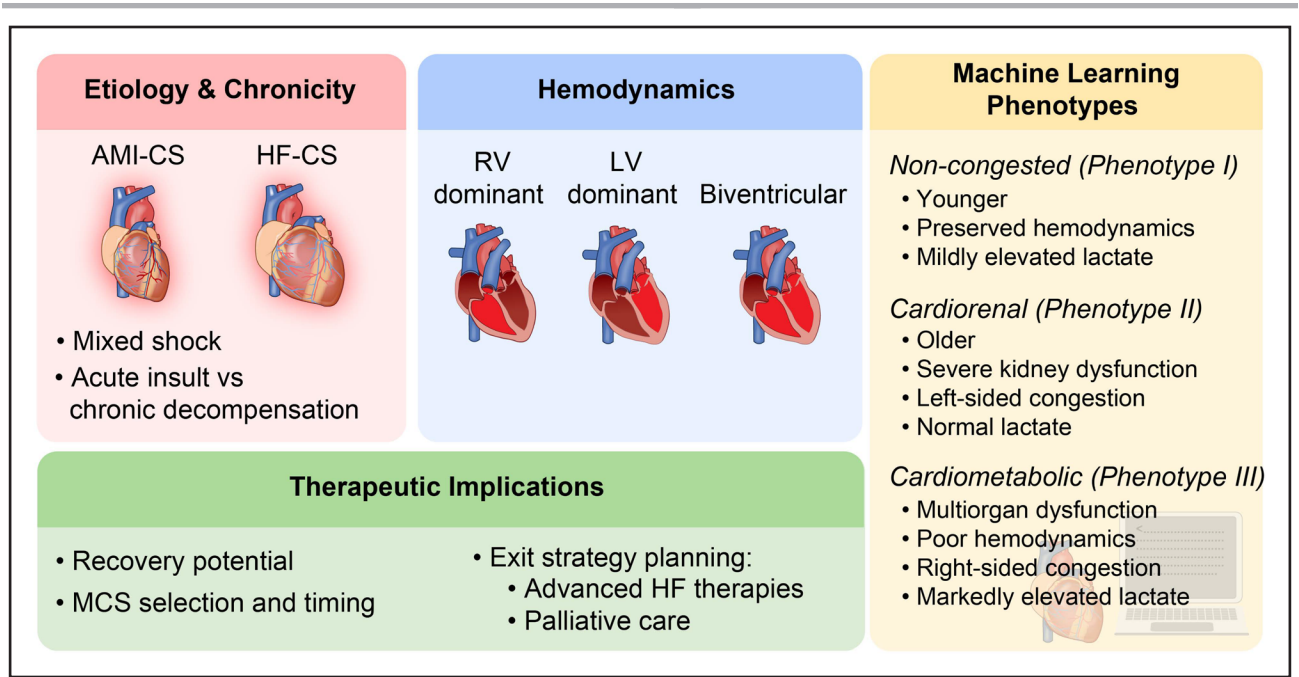


Figure 3. Clinical and hemodynamic phenotyping: one-size might not fit all. Phenotyping encompasses cause and chronicity of cardiogenic shock, hemodynamic classification, and machine-learning clusters. AMI-CS indicates acute myocardial infarction-related cardiogenic shock; HF-CS, heart failure-related cardiogenic shock; LV, left ventricular; MCS, mechanical circulatory support; and RV, right ventricular.

CS pathophysiology and highlight the importance of continuous reassessment rather than static diagnostic categorization. Temporal context further refines phenotyping. Patients with acute transient myocardial injury (eg, ischemic, inflammatory, toxic insults) often retain greater intrinsic capacity for functional recovery.⁶⁴ In contrast, individuals with longstanding HF often exhibit diminished metabolic flexibility, impaired mitochondrial energetics, and extensive structural remodeling characterized by myocyte loss, interstitial fibrosis, and ventricular dilation.⁶⁵ These changes reduce myocardial efficiency, contractile reserve, and structural reversibility, limiting the ability of the heart to recover.⁶⁵

Differentiating between acute, potentially reversible injury and chronic decompensation has critical prognostic and therapeutic implications. Patients with potentially reversible etiologies may benefit from aggressive early support, including temporary MCS as a bridge to recovery. Conversely, those with chronic, advanced cardiomyopathy may require earlier consideration of durable MCS or heart transplantation.

Hemodynamic Classification

Hemodynamic assessment classifies patients with CS into congestion profiles: LV-dominant, RV-dominant, or biventricular.³ LV-dominant CS is characterized by pulmonary capillary wedge pressure or LV end-diastolic pressure >15 mmHg with relatively normal right-sided filling pressures.^{3,66} RV-dominant CS is characterized

by relatively normal left-sided filling pressures with elevated right atrial pressure >15 mmHg, a right atrial to pulmonary capillary wedge pressure ratio >0.6, and a low pulmonary artery pulsatility index (<0.9 for AMI-CS, <1.85 in HF patients receiving durable MCS).^{3,66} Biventricular CS is marked by elevation in both right and left-sided filling pressures, and has been associated with worse outcomes and more frequent need for advanced HF therapies in studies using multicenter registry and administrative claim data.^{3,38,67}

Defining the congestion and failure phenotype is critical for MCS device selection, with intraaortic balloon pump (IABP) typically used for LV support, different percutaneous ventricular assist device (PVAD) modalities for LV or RV support, and veno-arterial extracorporeal membrane oxygenation (VA-ECMO) used for biventricular and respiratory support. Serial assessment is essential to assess the evolving physiology, adequacy of ventricular unloading, readiness for de-escalation, and likelihood of myocardial recovery.^{3,66}

Machine Learning Phenotypic Clustering

The Cardiogenic Shock Working Group developed a machine learning-based approach to phenotype patients with CS at admission, using readily available clinical parameters including age, sex, white blood cell count, platelet count, serum bicarbonate, kidney and liver function, and lactate levels.⁶¹ This analysis identified and validated three distinct phenotypes:

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noncongested (phenotype I), cardiorenal (phenotype II), and cardiometabolic CS (phenotype III).^{60,61} The noncongested phenotype included younger patients with fewer comorbidities, relatively preserved hemodynamics, and mildly elevated lactate levels. The cardiorenal phenotype included older patients with severe kidney dysfunction, left-sided (pulmonary) congestion, and normal lactate levels. The cardiometabolic phenotype encompassed patients with the highest illness severity, marked by multiorgan dysfunction, poor hemodynamics, right-sided (systemic) congestion, and markedly elevated lactate levels. These phenotypes demonstrate independent prognostic value when integrated with SCAI staging, with in-hospital mortality rates of 23%, 41%, and 52% for phenotypes I, II, and III, respectively.⁶⁰

Moreover, these phenotypes have been associated with distinct trajectories and differential responses to MCS, with implications for myocardial recovery potential and tailored treatment strategies. Most patients exhibited transitions in SCAI stages and phenotypes within the first six hours of admission, and remained relatively unchanged throughout the next 72 hours.⁶⁸ Phenotypes II and III have been associated with a significantly higher probability of progression to SCAI stage D or E or death within 72 hours of admission.⁶⁸ From an MCS standpoint, IABP use has been associated with the lowest mortality in patients with cardiorenal CS, PVAD with the lowest mortality in noncongested and cardiometabolic CS, with VA-ECMO being associated with the highest mortality across phenotypes.⁶⁰ In a secondary analysis of the DanGer trial, noncongested patients appeared to derive the greatest benefit from PVAD.⁶⁹ Although these data are observational and prospective validation is needed, phenotype-specific treatment algorithms could help improve recovery rates and CS outcomes.

TARGETED THERAPIES TO PROMOTE MYOCARDIAL RECOVERY

Optimization of Conventional Pharmacotherapies

Vasoactive and inotropic therapies provide critical hemodynamic support during the acute phase of CS, yet their role in promoting myocardial recovery remains equivocal (Figure 4).^{2,19} While they indirectly facilitate recovery as a bridge to reversal of the underlying pathology through ventricular unloading and use of disease-modifying therapies, their use should be carefully balanced against adverse effects, including increased myocardial oxygen consumption, arrhythmogenesis, and myocardial injury.^{2,70}

Vasoactive agent selection influences outcomes in CS, with norepinephrine emerging as the preferred first-line agent based on comparative trial data.^{2,71} The SOAP

II trial (Sepsis Occurrence in Acutely Ill Patients II) demonstrated lower rates of tachyarrhythmias with norepinephrine compared with dopamine in patients with CS.⁷² The OptimaCC trial (Epinephrine Versus Norepinephrine for Cardiogenic Shock After Acute Myocardial Infarction) was terminated early due to a significantly higher incidence of refractory AMI-CS in patients receiving epinephrine versus norepinephrine.⁷³ Current guidelines recommend titrating vasopressors to the lowest possible dose to sustain perfusion, minimizing undesired adrenergic effects that could compromise myocardial recovery.^{3,19} Dobutamine and milrinone are the most frequently used inotropes.² The DOREMI trial (Milrinone as Compared with Dobutamine in the Treatment of Cardiogenic Shock) randomized 192 patients with CS (94% in SCAI stages C–E) to milrinone versus dobutamine and found a nonsignificant difference in the primary composite outcome of in-hospital death, resuscitated cardiac arrest, heart transplantation or MCS, nonfatal AMI, stroke, or renal replacement therapy.⁷⁴ Secondary outcomes, including in-hospital mortality, lactate clearance, and urine output, also did not differ.⁷⁴ Milrinone offers theoretical advantages, including greater reductions in filling pressures through potent vasodilation, less tachycardia, and preserved efficacy in patients on beta-blockers, but is renally cleared, has a longer half-life, and carries the risk of hypotension.^{2,4,71} Levosimendan, a calcium sensitizer with inotropic and vasodilatory properties, may limit some catecholamine-mediated toxicity, but randomized evidence is lacking.^{4,71}

Strategies of permissive hypotension by accepting mean arterial pressure <65 mmHg, integration of phenotype-specific approaches such as use of inodilators in normotensive patients with elevated systemic vascular resistance, avoiding milrinone in patients with renal dysfunction, and earlier consideration of MCS as opposed to pharmacotherapy intensification, have been supported by observational data and proposed in CS expert consensus statements as reasonable strategies in the management of patients with CS.^{2,3,75–78} These strategies introduce a plausible mechanistic basis for minimizing myocardial injury and facilitating recovery. In addition, the use of intravenous or oral inotropes as a bridge to initiation or uptitration of guideline-directed HF pharmacotherapy provides theoretical advantages in promoting myocardial recovery, by counteracting the potential negative effects of inotropic agents alone. This concept was proposed in the EMPOWER study (Enoximone Plus Extended-Release Metoprolol Succinate in Subjects With Advanced Chronic Heart Failure) (REGISTRATION: URL: <https://clinicaltrials.gov/study/NCT00077948>; Unique identifier: NCT00077948), which proposed the use of oral phosphodiesterase inhibition to facilitate beta-receptor blockade in patients with advanced HF.⁷⁹ Although the trial was ultimately not conducted, the principle of pharmacological support to enable

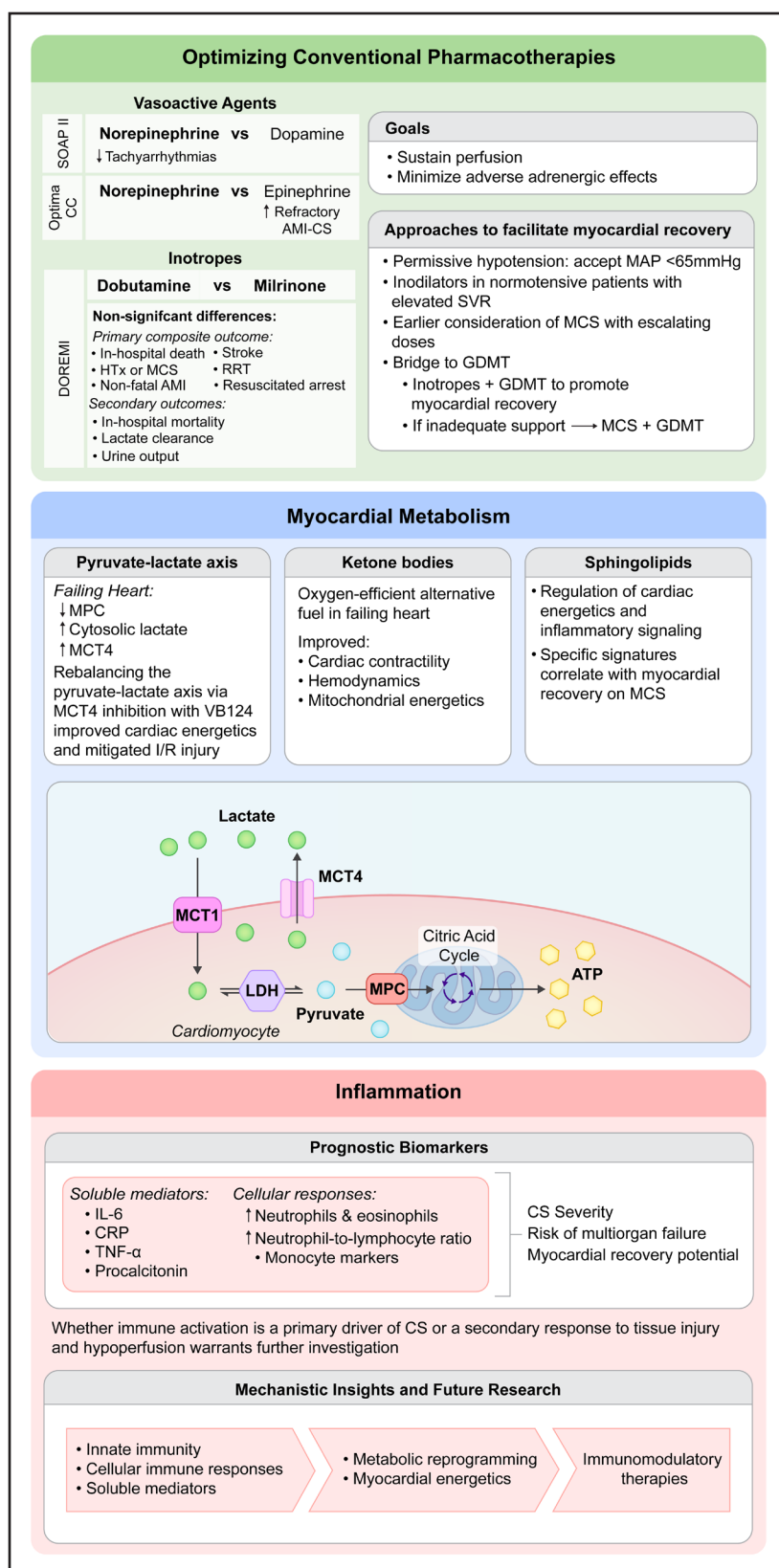


Figure 4. Targeted therapies to promote myocardial recovery. Optimization of conventional pharmacotherapies and potential therapies targeting myocardial metabolism and inflammation. AMI indicates acute myocardial infarction; CRP, C-reactive protein; CS, cardiogenic shock; GDMT, guideline-directed medical therapy; HF-CS, heart failure-related cardiogenic shock; HTx, heart transplantation; IL-6, interleukin-6; I/R, ischemia/reperfusion; LDH, lactate dehydrogenase; MAP, mean arterial pressure; MCS, mechanical circulatory support; MCT1, monocarboxylic anion transporter-1; MPC, mitochondrial pyruvate carrier; RRT, renal replacement therapy; SVR, systemic vascular resistance; and TNF- α , tumor necrosis factor- α .

disease-modifying therapy remains clinically relevant. In patients in whom pharmacological support is insufficient or poorly tolerated, temporary MCS may provide a more

effective platform for guideline-directed HF therapies and facilitate myocardial recovery. This strategy is discussed in detail in a dedicated section below.

Myocardial Metabolism

Under normal conditions, the heart primarily relies on fatty acid oxidation (FAO; 50% to 70%) for energy production, with pyruvate contributing an additional 20% to 40% via glycolysis and lactate conversion.^{80,81} With increased metabolic demand, the myocardial uptake of lactate and its use as a metabolic fuel increase.^{80,81} Lactate enters cardiomyocytes via the monocarboxylic anion transporter (MCT), is converted into pyruvate by lactate dehydrogenase, and then pyruvate enters the mitochondria via the mitochondrial pyruvate carrier to produce ATP, with a higher ATP yield compared with FAO.⁸⁰ Mitochondrial pyruvate carrier, which likely represents a key cellular decision point between pyruvate oxidation versus cytosolic conversion to lactate, is highly expressed in the healthy adult heart, while MCT4, which is believed to be the primary lactate exporter, is expressed at low levels.^{82–86} In the chronically failing heart, there is a shift from FAO to glycolytic metabolism, compromising the ability of cardiomyocytes to produce sufficient ATP and leading to an energy-deficient state.⁸⁰ Mitochondrial pyruvate carrier expression is downregulated, the conversion of pyruvate to lactate increases, and cytosolic lactate accumulates.⁸⁰

The impact of this metabolic shift on the acutely or chronically failing heart remains a matter of debate.⁸⁷ Elevated lactate levels have been shown to activate sodium-calcium exchangers and negatively impact cardiac function.^{88,89} Emerging evidence, however, suggests that lactate may serve as a crucial energy substrate in HF and CS.^{90–93} Lactate infusion has been reported to increase cardiac output and LV ejection fraction in animal models,^{94,95} with similar effects reported in a clinical trial of patients with CS.⁹⁶ MCT4 expression increases in response to the hypoxic and glycolytic conditions present in HF through a hypoxia-induced factor-1 α -dependent mechanism, enabling increased lactate efflux from cardiomyocytes that have shifted toward glycolytic metabolism.^{83,97} Pharmacological MCT4 inhibition improved cardiac function and ameliorated pathological damage.⁹⁸ In ischemia-reperfusion injury, which is often the underlying pathophysiology driving acute HF and CS after AMI, our team inhibited MCT4 with a new small-molecule drug (VB124) in a mouse model. This strategy normalized reactive oxygen species, mitochondrial membrane potential, and Ca²⁺ levels; increased pyruvate entry to the Krebs cycle; increased oxygen consumption; and improved myocardial salvage and functional outcomes.⁹⁹ These data suggest that normalizing pyruvate-lactate metabolism by inhibiting MCT4 is a promising therapy to mitigate ischemia-reperfusion injury and improve myocardial recovery outcomes in AMI-CS (Figure 4).

Ketone bodies serve as an oxygen-efficient alternative fuel substrate for the failing heart.⁸⁰ Ketone oxidation

requires less oxygen per ATP generated compared with fatty acids, making it metabolically advantageous during the hypoxic conditions of CS.⁸⁰ In HF with reduced ejection fraction, ketone oxidation contributes $\approx$ 16.4% of myocardial ATP production, compared with only 6.4% in healthy hearts.¹⁰⁰ This metabolic shift is proportional to circulating ketone concentrations, indicating that ketone availability determines cardiac utilization.⁸⁰ In a porcine model of ischemic CS, infusion of 3-hydroxybutyrate enhanced LV contractility, as demonstrated by increased end-systolic elastance.⁹⁵ In the KETO-SHOCK trial (Treatment With the Ketone Body 3-hydroxybutyrate in Patients With Cardiogenic Shock), ketone ester administration led to improvements in LV ejection fraction and cardiac power output and reductions in both left- and right-sided filling pressures.¹⁰¹ Infusion of 3-hydroxybutyrate augmented respiratory efficiency in isolated cardiac mitochondria and increased the cytosolic phosphocreatine/ATP ratio, indicating an improved energy status.⁹⁵ This mitochondrial protection may be particularly important in preventing further deterioration and promoting myocardial recovery (Figure 4).

Whether the myocardium can be therapeutically redirected to re-engage FAO as a primary ATP source remains an area of active investigation. While restoring FAO may improve energetic efficiency, excessive lipid uptake in the absence of coordinated oxidation can exacerbate lipotoxicity, underscoring the need for balanced metabolic modulation.¹⁰² In this context, sphingolipid metabolism has emerged as a critical regulator of both cardiac energetics and inflammatory signaling. Circulating and myocardial sphingolipids, including ceramides and sphingosine-1-phosphate, have been studied in patients with HF and MCS.^{103–105} Elevated ceramide species have been associated with mitochondrial dysfunction, impaired FAO, and adverse remodeling, whereas sphingosine-1-phosphate appears to exert cardioprotective effects through pro-survival and anti-inflammatory signaling pathways.^{106–111} Recent studies suggest that specific sphingolipid signatures correlate with myocardial recovery following MCS, highlighting a potential link between lipid remodeling, immune activation, and metabolic reprogramming (Figure 4).¹¹²

The above suggests a potential therapeutic role for metabolism modulation in CS and may represent a promising future direction. Although these emerging metabolic pathways are supported by preclinical studies and early clinical observations, further mechanistic and translational research is needed to elucidate the underlying mechanisms and determine whether they can be effectively leveraged in clinical practice. The platform of MCS-mediated myocardial recovery provides a unique opportunity to investigate the molecular changes of cardiomyocytes before and after MCS, with a particular focus on myocardial energetics.¹¹³

Inflammation

Immune and inflammatory responses play a central role in the pathophysiology of CS.^{2,114–117} Multiple clinical studies, predominantly in the setting of AMI-CS, have investigated circulating cytokines and inflammatory mediators as both mechanistic contributors and prognostic biomarkers.^{118,119} IL-6 (interleukin-6), C-reactive protein, tumor necrosis factor- α , and procalcitonin were predictive of short-term mortality, with IL-6 emerging as a marker and mediator of systemic inflammation, with levels correlating with CS severity, extent of myocardial injury, risk of multiorgan failure, and myocardial recovery potential.^{114,116,117,120,121} CS is increasingly recognized as a systemic inflammatory syndrome, as evidenced by the frequent occurrence of sepsis-induced cardiomyopathy.^{62,63} Current literature on immunity in CS remains heavily weighted toward AMI-CS and is largely based on observational associations and prognostic correlations, rather than direct mechanistic interrogation.^{114–117} While these studies clearly demonstrate that inflammatory pathways are clinically relevant, whether immune activation is a primary driver of CS or a secondary response to tissue injury and hypoperfusion remains unsettled.^{2,62} Furthermore, the upstream triggers, cellular sources, and regulatory checkpoints governing this immune response, particularly the interplay between innate immunity, metabolic reprogramming, and myocardial energetics, remain incompletely defined. Addressing this gap will be essential to move beyond biomarker-based risk stratification and toward targeted immunomodulatory therapies. While interleukin-1 and IL-6 blockade in AMI seemed beneficial, interventions in CS have been limited to extracorporeal cytokine removal, and, as such, further research is warranted.^{121–124}

Beyond soluble mediators, recent studies have investigated cellular immune responses, demonstrating dynamic and time-dependent alterations in circulating immune cell populations.^{115,125} Elevated neutrophil and eosinophil counts have been described in early CS, while the neutrophil-to-lymphocyte ratio has been associated with survival.^{115,125} Distinct monocyte markers observed in AMI-CS compared with uncomplicated AMI further suggest that immune mechanisms contribute to the development of CS.¹²⁶ These associations are consistent with the broader body of cardio-immunology research, connecting immune cells to incident HF.^{127,128} Whether through the release of cytokines and chemokines, or mechanisms ranging from neutrophil extracellular traps to cardiomyocyte apoptosis, inflammation seems to markedly impact cardiac function.^{129–132} Analogous mechanisms are observed in other types of shock, where cytokine-driven inflammation and immune dysregulation promote refractory systemic vasodilation, vascular leakage, and circulatory failure.^{133,134}

One of the next frontiers in CS research is to link immune-mediated mechanisms with the clinical course

and heterogeneity of CS subtypes. A key objective is to define how these pathways evolve over time and how they are modified by pharmacotherapy, MCS, and immunomodulatory therapies. This approach could allow for precision interventions tailored to distinct pathophysiological profiles, and ameliorate or reverse cardiac and end-organ dysfunction (Figure 4).

TEMPORARY MCS AS A RECOVERY PLATFORM: FROM SALVAGE TO NATIVE HEART SURVIVAL

Temporary MCS should be viewed not only as a bridge to decision or advanced HF therapies but as an active platform to facilitate myocardial and end-organ recovery.^{66,77} In a recovery-oriented approach, overlapping phases of salvage, optimization, stabilization, and de-escalation can be used to define its role (Figure 5).³ During salvage, MCS aims to augment perfusion and limit further end-organ injury.^{3,78} During optimization, invasive hemodynamic profiling helps identify the dominant CS phenotype and align support with cardiac unloading and systemic perfusion targets.^{3,66} During stabilization, emphasis shifts to prevention and mitigation of secondary injury due to device-related complications, and reassessment of tolerance to disease-modifying therapies.^{3,135} During de-escalation, weaning functions as a test for cardiac recovery, with serial assessment of hemodynamics, end-organ function, and cardiac performance to determine readiness for explantation, consideration of advanced HF therapies, or transition to palliative care.^{66,136}

Salvage Phase: Indications and Timing

Temporary MCS is typically used when hypoperfusion persists despite initial pharmacological therapy, or systemic perfusion needs to be maintained while underlying CS pathologies are being treated.^{19,78} Major contraindications include irreversible cardiac or end-organ injury that precludes meaningful recovery. In practice, it seems that these decisions are best made by a multidisciplinary team.^{19,23,66,77} Our institution pioneered the concept of a multidisciplinary shock team approach in the management of CS.²⁵ The timing of MCS deployment has been shown to affect outcomes in observational studies, with delayed escalation to MCS potentially allowing for progressive hemo-metabolic deterioration, thus reducing the likelihood of recovery and NHS.^{7,8,76–78} This is especially relevant for VA-ECMO, where observational data suggest that outcomes vary substantially by underlying cause, with improved survival reported in acute reversible syndromes such as myocarditis, and poor outcomes in the setting of extracorporeal cardiopulmonary resuscitation and refractory vasoplegia.^{137,138}

Before initiation of MCS, the care team should explicitly address a few questions that define both the immediate

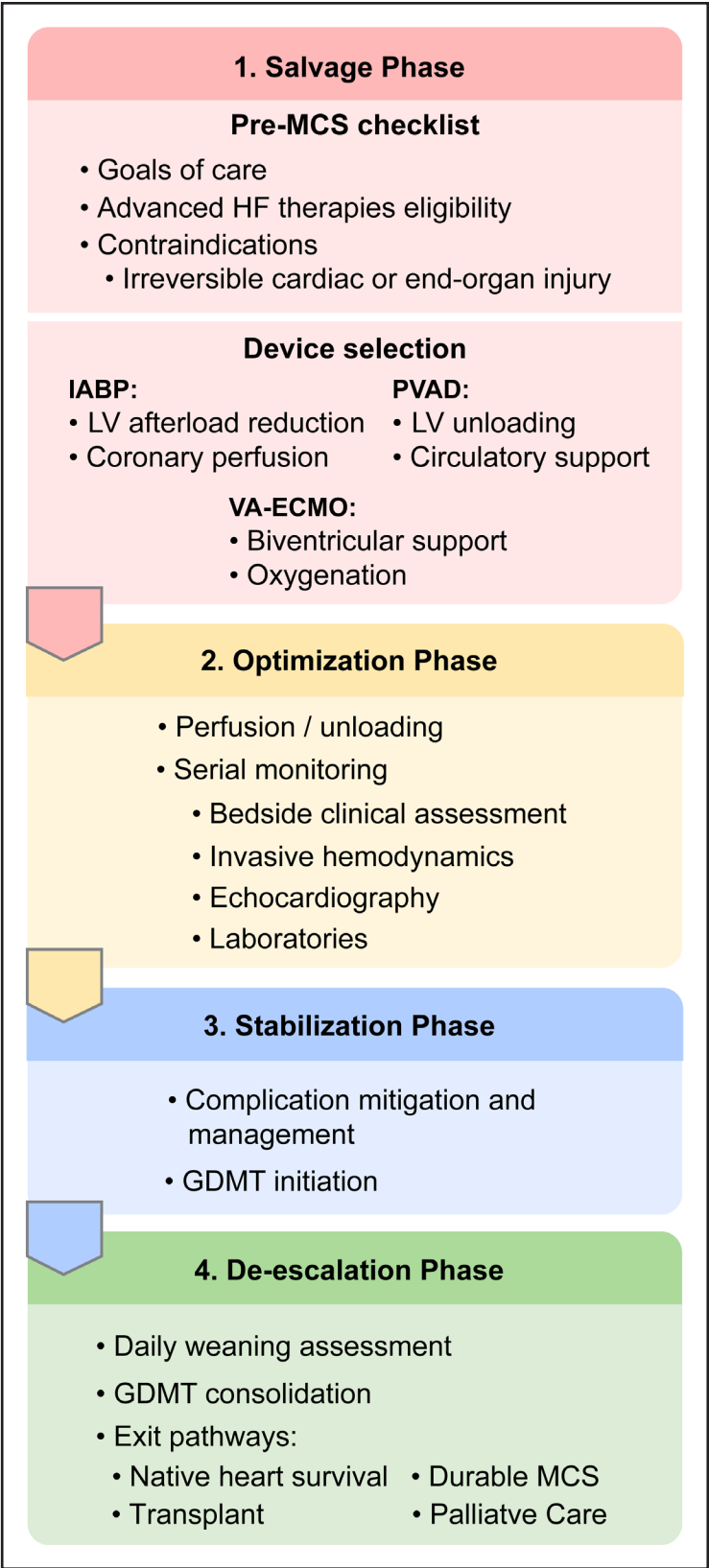


Figure 5. Temporary mechanical circulatory support as a recovery platform: From salvage to native heart survival. Incorporation of temporary circulatory support into the 4 proposed phases for the management of cardiogenic shock: Salvage, Optimization, Stabilization, and De-escalation. GDMT indicates guideline-directed medical therapy; HF, heart failure; IABP, intraaortic balloon pump; LV, left ventricular; MCS, mechanical circulatory support; PVAD, percutaneous ventricular assist device; and VA-ECMO, veno-arterial extracorporeal membrane oxygenation.

therapeutic goals and the downstream exit strategy.⁶⁶ First, what problem are we aiming to address? The dominant hemodynamic and metabolic derangements should be identified. Is the primary problem inadequate perfusion, inadequate unloading, RV failure, respiratory failure, or a mixed phenotype? Device selection should be guided by the dominant phenotype rather than applied through a one-size-fits-all approach.^{3,66} Second, what are the intended clinical targets? Specific hemodynamic and metabolic goals should be defined upfront, and in more severe or refractory cases, higher-flow platforms may be required.^{3,66} Third, what is the exit strategy? Before device insertion, the team should define whether temporary MCS is intended as a bridge to recovery, bridge to decision, bridge to advanced HF therapies, or whether MCS is unlikely to bridge to a meaningful clinical goal.⁶⁶ Early goals of care discussion and palliative care consultation should be considered.¹³⁹

Optimization Phase: Hemodynamics, Unloading, and Multimodal Monitoring

Once MCS is deployed, management shifts from initial rescue to phenotype-guided optimization.^{66,140} The goal is an optimal balance between perfusion and unloading that supports end-organ recovery and limits cardiac stress.^{66,140} This is reflected by improvement in filling pressures, cardiac output, perfusion pressure, and end-organ function.^{66,141} In a retrospective, multicenter study of patients receiving PVAD, only 37% met combined 24-hour hemodynamic and metabolic goals, and failure to do so was associated with a 3.2-fold higher risk of 30-day mortality or MCS reintroduction.¹⁴¹ Multicenter registry data and a post hoc analysis of the HYPO-ECMO trial (Effects of Induced Moderate Hypothermia on Mortality in Cardiogenic Shock Patients Rescued by Veno-arterial ExtraCorporeal Membrane Oxygenation) suggested that with VA-ECMO, lactate clearance and improved pulse pressure over the first 24 hours may indicate improving cardiac function rather than complete dependence on extracorporeal support.^{142,143}

Because unloading profiles differ substantially across devices, support should be tailored to the dominant physiology.^{77,140} IABP employs a counterpulsation mechanism to increase diastolic blood pressure and decrease LV afterload and myocardial oxygen consumption, leading to modest increases in cardiac output and LV unloading.^{77,144} PVAD propels blood from the LV into the aorta and thereby augments cardiac output and decreases LV filling pressures and wall stress.^{77,145} It reduces LV work and oxygen demand, creating a hemodynamic environment that facilitates recovery.^{140,145} VA-ECMO provides potent circulatory and respiratory support but in a proportion of patients, can increase LV afterload through retrograde blood flow, leading to LV distension, higher LV end-diastolic pressure, impaired aortic valve opening, and

intracardiac stasis.^{77,146} However, it needs to be emphasized that these effects are not universal. In the ECMO-Flow trial (Effect of ECMO Flow Variations on Pulmonary Capillary Wedge Pressure in Patients With Cardiogenic Shock), stepwise increases in VA-ECMO flow from 2 to 4 L/min resulted in stable or decreased pulmonary capillary wedge pressure in 94% of patients.¹⁴⁷ The expected rise in LV filling pressures with increasing afterload appeared to be counterbalanced by reductions in RV preload.¹⁴⁷ Hence, the hemodynamic response to VA-ECMO seems to be more heterogeneous than previously assumed and is defined by patient-specific factors, including biventricular function and filling pressures and volume status. Nevertheless, when inadequate unloading and LV distension occur, recovery may be compromised if not recognized early and managed appropriately.¹⁴⁸ Strategies to mitigate this include venting with PVAD, IABP, atrial septostomy, or transseptal left atrial cannulation, together with adjunctive inotropes, decongestion and lowering of filling pressures with diuretics and vasodilators, and use of the lowest possible flow.^{77,146,149} Observational studies and meta-analyses suggest that LV unloading during VA-ECMO may be associated with lower mortality.^{150–152} However, early routine transseptal unloading did not confirm this benefit in the EARLY UNLOAD randomized trial (Early Left Atrial Septostomy Versus Conventional Approach After Venoarterial Extracorporeal Membrane Oxygenation).¹⁵³ Potential benefits should be weighed against increased device-related complications and noninvasive approaches should also be tested.^{148,151}

In the absence of randomized data and protocols to guide hemodynamic optimization during MCS, current management approaches integrate data derived from bedside clinical assessment, invasive hemodynamics, echocardiography, and laboratory trends.^{66,154,155} This multimodal approach aims to assess whether support is merely maintaining circulation or facilitating cardiac improvement and reversal of end-organ injury.^{66,155} Serial echocardiography assesses ventricular performance, valve function, and device position, while laboratory assessment of lactate, renal and hepatic markers, and mixed venous oxygen saturation provides complementary information about systemic perfusion.^{66,154} Perfusion pressure targets should be tailored to patient symptoms and end-organ response, while maintaining low afterload, rather than applied as universal thresholds.^{29,66,156} Interpretation requires recognition of the limitations of standard hemodynamic metrics during MCS. With VA-ECMO, estimates of cardiac output via Fick or thermodilution may not reflect total effective systemic flow, while continuous-flow support might make pulse pressure a less reliable marker of ventricular performance.^{35,66,146} In these settings, clinical trajectory, laboratory assessment, and cardiac imaging become particularly important in guiding ongoing support and readiness for de-escalation.^{66,154}

Stabilization Phase: Complication Mitigation and Bridge to HF Therapies

Once hemodynamic goals are achieved, the stabilization phase shifts the focus to active recovery enablement.^{66,140} The priority is to prevent secondary injury while consolidating the gains achieved with MCS through treatment of the underlying cause, decongestion, and reassessment of tolerance to the introduction of guideline-directed HF medical therapies.^{3,66} As detailed later in this section, this strategy will be tested in the prospective, multicenter INTEGRATE study (Initiation and Termination/Titration of Guideline Directed Medical Therapy in Heart Failure Cardiogenic Shock With Impella 5.5 for Cardiac Recovery) (REGISTRATION: URL: <https://clinicaltrials.gov/study/NCT06965504>; Unique identifier: NCT06965504). In this sense, stabilization is not a passive holding phase, but a period in which hemodynamic rescue evolves into durable clinical improvement.

Complication surveillance is central as the benefits of MCS can be rapidly offset by secondary injury.^{66,77} Adverse events during MCS are broadly vascular, neurological, hematologic, mechanical, and infectious, and their frequency varies by device type and patient characteristics.⁶⁶ Devices requiring larger-bore vascular access are associated with greater risks of bleeding and vascular complications, whereas hemolysis is particularly relevant with PVAD.^{66,157–159} These events are not merely procedural, as they may prolong shock, delay recovery, increase resource use, and complicate de-escalation.^{157,160} Hemolysis often reflects device malposition or decreased preload, prompting repeat imaging and volume status assessment.^{66,158,159} Bleeding risk is amplified by anti-thrombotic therapy and device-mediated hemostatic abnormalities, requiring careful balance between hemorrhagic and thrombotic risk.^{140,158,159} In patients on VA-ECMO, attention to distal limb perfusion, aortic valve opening, LV distention, and upper body oxygenation is especially important.^{66,146,160} Complication mitigation should be viewed as a core component of recovery-oriented care rather than a secondary maintenance task.

The stabilization phase gives the opportunity for longer-term disease modification.^{3,66} Temporary MCS might help support perfusion while HF pharmacotherapy is introduced.^{3,5,161,162} Most HF medical therapy trials excluded patients with CS, however, observational data suggest that combining MCS with medical therapy might be associated with improved outcomes.^{5,161,162} The INTEGRATE prospective, single-arm, multicenter study (Initiation and Termination/Titration of Guideline Directed Medical Therapy in Heart Failure Cardiogenic Shock With Impella 5.5 for Cardiac Recovery) aims to assess whether PVAD support can facilitate HF pharmacotherapy optimization in patients with HF-CS (REGISTRATION: URL: <https://clinicaltrials.gov/study/NCT06965504>; Unique identifier: NCT06965504). In the absence of

prospective data to guide HF pharmacotherapy introduction, the timing and sequence vary and are typically guided by hemodynamic stability and level of MCS, end-organ recovery, and degree of congestion.^{135,156} Mineralocorticoid receptor antagonists and sodium-glucose cotransporter-2 inhibitors are usually incorporated first, given their lesser hemodynamic effects and anticongestive properties, followed by renin-angiotensin system inhibitors, with close monitoring of renal function and serum electrolytes, while beta-blockers are usually introduced after establishing hemodynamic stability with a decreased level of MCS, given their negative inotropic and chronotropic effects.^{5,19,156,161} The broader objective during this phase is the consolidation of recovery by coupling hemodynamic stabilization with definitive treatment of the underlying cause, aggressive decongestion, and institution of disease-modifying therapies.^{3,19,156}

De-Escalation Phase: Weaning Strategies and Exit Pathways

The de-escalation phase describes the transition from MCS dependence to weaning, with serial assessment of cardiac reserve.^{66,136} MCS weaning should be a structured recovery trial to determine whether the patient can sustain perfusion or whether to instead consider advanced HF therapies or transition to palliation.^{3,66} Although high-quality evidence remains limited, a pragmatic approach includes daily reassessment, stepwise reduction in support, and multimodal monitoring to ensure stability.^{3,66,136}

The optimal timing of weaning is not well-defined in the absence of prospective data and guidelines, with expert consensus statements suggesting that weaning might be reasonable to begin with evidence of reversal of end-organ hypoperfusion, euvoemia, minimal inopressor requirements, and improved cardiac performance on echocardiography.^{3,66} The precipitating cause of shock should have been reversed or clearly improving, and major device-related complications should be resolved or stable.^{3,66} Daily weaning assessment is recommended and should proceed through a stepwise, device-specific reduction in support, with the pace determined by the reversibility of the underlying insult and the patient's hemodynamic and metabolic response.^{3,66} This is reflected by acceptable filling pressures and stable mean arterial pressure, lactate, and hepatic and renal function, with individualized interpretation in patients with chronic HF.^{66,136}

IABP weaning is typically performed by ratio wean (gradual reduction in counterpulsation frequency) or, less commonly, volume wean (progressive balloon deflation), depending on device platform and local practice.^{66,136,163} A randomized trial of 30 patients supported with IABP suggested better hemo-metabolic indices with volume wean compared with ratio wean.¹⁶³ Several PVAD weaning protocols have been described that entail a gradual

reduction in the level of support with serial monitoring of LV function and size, hemodynamics, and end-organ perfusion.^{3,136} With cardiac recovery, total cardiac output should be maintained despite a reduction in pump speed, and stability should be maintained before explantation.¹³⁶ VA-ECMO weaning typically involves gradual pump speed reduction by 0.5 to 1 L/min decrements to 1.5 to 2 L/min, with some centers transiently reducing to 1 L/min under echocardiographic guidance.^{66,136} Explant prerequisites as per expert guidance include improved LV size and function, aortic valve opening with each cardiac cycle, and improved valvular regurgitation at minimal ECMO flow.¹³⁶ In patients supported with combined VA-ECMO and PVAD, VA-ECMO is typically weaned first while maintaining PVAD to preserve unloading.^{66,136} If support reduction leads to persistent hemodynamic deterioration or clinical decompensation, the trial should be stopped and support restored to the last stable setting.^{66,154}

Because unsuccessful weaning is not uncommon, the potential exit pathways after temporary MCS should be discussed before MCS deployment.^{3,66} In patients who cannot be weaned and are not heart transplantation candidates, or who prefer MCS as a destination therapy, transition to durable MCS is an established pathway that can improve survival, quality of life, and functional capacity.^{19,164} The recovery intent need not end with durable MCS, as an a priori bridge-to-recovery strategy has been associated with a greater likelihood of recovery and subsequent explantation in an analysis of consecutive patients with LVAD enrolled in the INTERMACS registry (Interagency Registry for Mechanically Assisted Circulatory Support), and tools such as the INTERMACS cardiac recovery score and the multicenter RecoverHeart calculator might aid with decision-making.^{13,16,164,165} Heart transplantation remains an important pathway for selected patients supported with temporary MCS, although candidacy and outcomes may be constrained by critical illness severity and multiorgan dysfunction.^{2,166} When recovery is not achievable and advanced HF therapies are either not appropriate or not desired, transition to palliative care becomes the appropriate exit strategy.^{3,66} Ideally, these options should be evaluated early, revisited throughout the period of support, and aligned with patient values and multidisciplinary clinical assessment.

CONCLUSIONS

Myocardial and end-organ recovery should be an explicit target rather than an optimistic outcome in the management of patients with CS. In a recovery-focused framework, attention shifts from acute rescue to long-term recovery facilitation. Early identification and triage, integration of macro- and microcirculatory monitoring, phenotype-guided therapies aimed at reducing cardiac

workload and reversing hemo-metabolic stress, followed by structured de-escalation and implementation of disease-modifying therapies, aim to preserve the reversibility of cardiac and peripheral organ injury and enable recovery and NHS. Within this continuum, temporary MCS should be viewed not only as a bridge to decision or advanced HF therapies, but as a recovery-enabling platform that, when used in a timely manner, matched to phenotype, and guided by integrated monitoring, can enable sustained myocardial and end-organ recovery. Future efforts should focus on establishing standardized, recovery-oriented end points, identifying clinical phenotypes and biomarkers suggesting reversible pathophysiology, implementing recovery-directed treatment strategies, and evaluating their impact on long-term clinical outcomes.

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